

Memo 09: Spatial Distribution Analysis for FVCOM-MP Cases C and D

Effective Floc Coupling, Days 10 and 30

FVCOM-MP WaterPACT Investigation

May 2026

Purpose

This memo records the first spatial-distribution comparison between case `c`, the comparison MP-floc case, and case `d`, the `ORIG_SED` effective-floc coupling case. The analysis used the day-10 and day-30 MAT files already extracted in `OUTPUT_c` and `OUTPUT_d`. Day 90 is not included here because the corresponding v2 MAT files were not present at the time of this memo.

Files and Command

The analysis inputs were:

- `FVCOM_MP_test_run/OUTPUT_c/spatial_distribution_v2_c_day10.mat`
- `FVCOM_MP_test_run/OUTPUT_d/spatial_distribution_v2_d_day10.mat`
- `FVCOM_MP_test_run/OUTPUT_c/spatial_distribution_v2_c_day30.mat`
- `FVCOM_MP_test_run/OUTPUT_d/spatial_distribution_v2_d_day30.mat`

The analysis was run with:

```
python FVCOM_MP_test_run/PYTHON/analyze_spatial_distribution_v2.py --days 10 30
```

The outputs were written to:

- `FVCOM_MP_test_run/PYTHON/output/spatial_distribution_v2/day10`
- `FVCOM_MP_test_run/PYTHON/output/spatial_distribution_v2/day30`

Each day folder contains per-case maps, `d-c` maps, `summary_stats.csv`, `diff_summary_stats.csv`, and `README.md`. The run generated 108 PNG maps across the two day folders.

Extraction Window

Both day-10 and day-30 MAT files contain 24 output records. The day-10 files span simulation day 9.0000 to 9.9570, and the day-30 files span simulation day 29.0000 to 29.9570. These are therefore final-24-hour averages before the target day endpoints, as intended for this v2 spatial workflow.

Equivalent Floc Size Post-processing

The equivalent floc size was computed from the existing output `settle_vel_floc_mp1` with

$$d_{\text{floc,eff}} = \left[\frac{18\mu W_{\text{floc}}}{g(\rho_{\text{floc,eff}} - \rho_w)} \right]^{1/2}, \quad (1)$$

where $\rho_{\text{floc,eff}} = 1300 \text{ kg m}^{-3}$ and

$$\nu = 10^{-6} \exp[-0.025(T - 20)], \quad 0.5 \times 10^{-6} \leq \nu \leq 2.0 \times 10^{-6}, \quad (2)$$

with $\mu = \rho_w \nu$. This is the same post-processing relation documented in Memo 08.

Key Spatial Statistics

Table 1 summarizes the domain means and 95th percentiles for the depth-averaged fields most relevant to the effective-floc coupling. All values are taken directly from the generated `summary_stats.csv` files.

Table 1: Depth-averaged domain statistics for selected fields.

Day	Field	C mean	D mean	C p95	D p95
10	Sediment	1.035	4.506	4.332	10.506
10	$W_{\text{floc}} \text{ (m s}^{-1}\text{)}$	1.463×10^{-3}	7.527×10^{-5}	3.000×10^{-3}	2.079×10^{-4}
10	$d_{\text{floc,eff}} \text{ (}\mu\text{m)}$	79.07	19.35	163.17	38.69
10	$\lambda_a \text{ (s}^{-1}\text{)}$	2.110×10^{-2}	0	8.713×10^{-2}	0
10	$\lambda_d \text{ (s}^{-1}\text{)}$	1.943×10^{-2}	4.996×10^{-1}	6.999×10^{-2}	5.000×10^{-1}
10	MP total (kg m ⁻³)	1.162×10^{-6}	6.697×10^{-7}	4.344×10^{-6}	2.700×10^{-6}
10	MP aggregated fraction	0.363	0	0.433	0
30	Sediment	0.524	4.373	2.518	10.143
30	$W_{\text{floc}} \text{ (m s}^{-1}\text{)}$	2.720×10^{-3}	1.174×10^{-4}	3.000×10^{-3}	2.809×10^{-4}
30	$d_{\text{floc,eff}} \text{ (}\mu\text{m)}$	148.31	30.89	167.08	47.62
30	$\lambda_a \text{ (s}^{-1}\text{)}$	1.028×10^{-2}	0	4.213×10^{-2}	0
30	$\lambda_d \text{ (s}^{-1}\text{)}$	8.798×10^{-3}	4.999×10^{-1}	3.934×10^{-2}	5.000×10^{-1}
30	MP total (kg m ⁻³)	3.495×10^{-5}	1.267×10^{-6}	1.702×10^{-4}	3.721×10^{-6}
30	MP aggregated fraction	0.181	0	0.388	0

Main Findings

The effective-floc settling signal changed strongly and plausibly

Case d has much smaller `settle_vel_floc_mp1` than case c. The domain-mean depth-averaged floc settling speed decreases from 1.46×10^{-3} to $7.53 \times 10^{-5} \text{ m s}^{-1}$ on day 10, and from 2.72×10^{-3} to $1.17 \times 10^{-4} \text{ m s}^{-1}$ on day 30. The maximum case-d depth-averaged values were 4.32×10^{-4} and $4.87 \times 10^{-4} \text{ m s}^{-1}$ on days 10 and 30, respectively, which is consistent with the ORIG_SED floc settling cap of approximately $5.0 \times 10^{-4} \text{ m s}^{-1}$.

The post-processed effective floc diameters are much smaller

The Stokes-equivalent floc size also decreases sharply in case **d**. The domain-mean depth-averaged effective diameter is about $19.4\ \mu\text{m}$ on day 10 and $30.9\ \mu\text{m}$ on day 30. The corresponding 95th percentiles are about $38.7\ \mu\text{m}$ and $47.6\ \mu\text{m}$. The maximum depth-averaged case-**d** values are approximately $61.6\ \mu\text{m}$ on day 10 and $66.0\ \mu\text{m}$ on day 30.

The present size gate disables aggregation in case **D**

The plastic input has `PLAST_AAXI` = `PLAST_BAXI` = `PLAST_CAXI` = 0.100, which is read as a 0.100 mm, or $100\ \mu\text{m}$, equivalent plastic diameter. In the current aggregation kernel, the critical MP size is computed from floc size as

$$d_{\text{MP,max}} = -0.0002d_{\text{floc}}^2 + 0.36d_{\text{floc}}, \quad d_{\text{floc}} < 900\ \mu\text{m}. \quad (3)$$

For the case-**d** effective floc range of roughly $20\text{--}66\ \mu\text{m}$, this gives $d_{\text{MP,max}}$ of only about $7\text{--}23\ \mu\text{m}$. The present $100\ \mu\text{m}$ MP class is therefore larger than the allowed size for aggregation. This explains why `lambda_a_mp1` is zero everywhere in case **d**, why `mp1_agg` is zero, and why the aggregated MP fraction is zero in both day-10 and day-30 outputs.

Disaggregation rate is high, but there is no aggregated MP mass

Case **d** has `lambda_d_mp1` close to $0.5\ \text{s}^{-1}$ across the domain. Because `mp1_agg` is zero in case **d**, this rate should be read mainly as a diagnostic rate produced by the kernel under the current effective-floc sizes, not as evidence of an active disaggregation flux.

Sediment fields differ substantially between cases

The case-**d** sediment concentration is higher in the depth average on both days: the mean is 4.51 versus 1.03 on day 10, and 4.37 versus 0.524 on day 30. Surface sediment is broadly higher in case **d**, while bottom sediment has mixed differences and can be lower than case **c**. This is expected to reflect the changed cohesive `ORIG_SED` sediment configuration in case **d**, not only the MP-side effective-floc switch.

MP response is strong

Case **d** has no aggregated MP and no bottom MP mass in the analyzed outputs. The domain-mean total suspended MP concentration is lower than case **c**: about 42% lower on day 10 and about 96% lower on day 30. This suggests that, under the present $100\ \mu\text{m}$ MP size and ORIG-equivalent floc sizes, the effective-floc coupling moves the model into a non-aggregation regime rather than a modified-aggregation regime.

Interpretation

The output confirms that the new post-processing workflow can see the intended case-**d** `ORIG_SED` settling signal through `settle_vel_floc_mp1`. The effective settling speed and the back-calculated equivalent floc size both change in a physically consistent direction: lower settling speed produces smaller Stokes-equivalent flocs.

The important model-behavior result is that these smaller effective flocs are too small to aggregate the current $100\ \mu\text{m}$ plastic class under the existing critical-size closure. Therefore, the case-**d**

run is not just changing the aggregation strength; it is effectively shutting aggregation off for this plastic class.

Sediment Setup Comparison and Equivalence Issue

The case-c and case-d results above should not be interpreted as an isolated test of only PLAST_FLOC_EFFECTIVE. The sediment configuration also changed between the two cases. Table 2 summarizes the active sediment-module inputs.

Table 2: Active sediment-module setup in cases c and d.

Parameter	Case c	Case d
Sediment file	<code>generic_sediment_c.inp</code>	<code>generic_sediment_d.inp</code>
SED_TYPE	<code>non-cohesive</code>	<code>cohesive</code>
SED_SD50	1.0 mm	0.004 mm
SED_WSET input	3.0 mm s ⁻¹	0.1 mm s ⁻¹
Runtime base settling speed	3.0×10^{-3} m s ⁻¹	1.0×10^{-4} m s ⁻¹
Settling treatment	constant non-cohesive Wset	ORIG_SED cohesive flocculation

In `mod_sed.F`, `SED_WSET` is read in mm s⁻¹ and converted to m s⁻¹. For non-cohesive sediment, the settling calculation uses `wset = sed(ised)%Wset`. For cohesive sediment, the code instead calls `Calc.Wset.Floc` with `sed(ised)%w0` as the base settling velocity and then applies the ORIG flocculation treatment. Therefore, case c uses a constant non-cohesive settling speed of 0.003 m s⁻¹, whereas case d starts from a cohesive base settling speed of 1.0×10^{-4} m s⁻¹ and then reconstructs the ORIG floc signal.

This means the current c–d comparison is not dynamically equivalent. Case d simultaneously changes:

- sediment type from non-cohesive to cohesive;
- primary sediment size from 1.0 mm to 0.004 mm;
- base settling speed from 3.0 mm s⁻¹ to 0.1 mm s⁻¹;
- the sediment settling closure from constant settling to ORIG cohesive flocculation;
- the MP interaction path through `PLAST_FLOC_EFFECTIVE=T`.

The plastic input files also contain a separate static parameter `SED_SD50 = 0.35` mm and `PLAST_WSSED = 0.01` m s⁻¹ in both cases. These are plastic-module ambient or bed parameters and are not the active suspended sediment class settings used by `mod_sed.F`. They should not be used to judge equivalence of the active sediment transport cases.

For a clean MP-coupling comparison, the control and switch-on cases should keep the active sediment module configuration identical. One practical setup would be:

1. define a cohesive control case with the same `SED_TYPE`, `SED_SD50`, `SED_WSET`, erosion, deposition, bed, and forcing settings as case d;
2. keep `PLAST_FLOC=T` but set `PLAST_FLOC_EFFECTIVE=F` in that control case;
3. compare it against a matched case with only `PLAST_FLOC_EFFECTIVE=T`.

That paired design would isolate the MP-side effective-floc-size treatment. The present case **c** versus case **d** comparison is still useful as a broad sediment-regime sensitivity, but it cannot support a strict attribution of the MP response to the effective-floc switch alone.

Recommended Next Checks

1. Confirm that the 0.100 mm MP size is the intended particle size for this sensitivity. If the intention is to test ORIG-effective floc aggregation, a smaller MP class may be needed.
2. Consider a sensitivity to `PLAST_FLOC_RHO_EFF`. Lowering the assumed effective floc density toward water density increases the Stokes-equivalent diameter for the same settling speed.
3. Consider adding optional diagnostics for $d_{\text{floc,eff}}$ and $d_{\text{MP,max}}$ directly in model output. The current post-processing can recover $d_{\text{floc,eff}}$, but $d_{\text{MP,max}}$ is still inferred from the aggregation closure.
4. Repeat the same v2 extraction and analysis for day 90 when the `spatial_distribution_v2-{c,d}_day90.mat` files are available.